



Khulna University of Engineering & Technology

Department of Computer Science and Engineering

Project Report

ResearchHub – Thesis Management Platform

Name	: Mutiur Rahman
Roll	: 220797
Year	: Third
Term	: First
Course No	: CSE 3100
Course Name	: Web Programming Laboratory
Date of Submission	: 20/07/2026

Abstract

Universities typically manage the thesis lifecycle through fragmented channels - email threads, shared drives, and messaging apps - resulting in lost documents, unclear deadlines, overwritten files, and no shared record of decisions between students and supervisors. ResearchHub is a centralized web application that addresses this by unifying the entire thesis workflow inside a single system: structured proposal submission and supervisor approval, automatic creation of a thesis workspace on approval, milestone and task tracking with dependency chains, versioned document uploads with SHA-256 checksums, meeting scheduling with optional Google Calendar/Meet integration, and threaded discussion with private supervisor notes.

The system is built on Laravel using server-rendered Blade views with Livewire for SPA-style navigation, and ships with a relational schema of thirteen application tables plus Laravel's standard authentication infrastructure, running on MySQL in development. Three roles are supported - Student, Supervisor, and Admin - each with a distinct dashboard and permission set enforced through role middleware and explicit ownership checks.

This report documents the system's architecture, database schema, and object model using Entity-Relationship and UML diagrams (class, use-case, sequence, and state diagrams), walks through its major functional modules, and presents annotated screenshots captured from a live, seeded instance of the running application.

Introduction

Supervising a thesis is a long, multi-stage collaboration between a student and a supervisor. In the absence of dedicated tooling, this collaboration fragments across email (proposal drafts and feedback), cloud drives (document versions, often with ad-hoc filenames such as "thesis_v3_final_FINAL.docx"), and messaging apps (meeting scheduling, quick questions). This causes three recurring problems: documents get overwritten or lost because there is no enforced version history; deadlines and milestones are tracked inconsistently or not at all; and there is no single, queryable record of what was agreed, when, and by whom.

ResearchHub was built to give a university department a single web application that every participant in the thesis lifecycle logs into, where the state of every proposal, thesis, milestone, document, and meeting is authoritative, timestamped, and access-controlled.

Objectives

- Provide a structured proposal submission and review workflow with explicit approve / reject / request-revision decisions.
- Automatically instantiate an active thesis workspace the moment a proposal is approved, eliminating manual project setup.
- Track progress via milestones and tasks, including dependency ordering and automatically computed completion percentages.
- Provide document management with immutable version history and integrity checksums, so no file is ever silently overwritten.
- Support meeting scheduling, including optional one-click Google Calendar / Google Meet link generation.
- Support threaded, role-aware discussion on each thesis, including supervisor-only private notes.
- Enforce role-appropriate access (Student / Supervisor / Admin) throughout, with a seeded demo dataset for evaluation.

Requirement Analysis

User Roles

The system defines exactly three roles via a PHP backed enum (UserRole: Student, Supervisor, Admin). Students and Supervisors may self-register; the first Admin is bootstrapped through a one-time /setup email-code flow, and additional admins are created only by promoting an existing Student or Supervisor from the admin user-management screen.

Permission Matrix

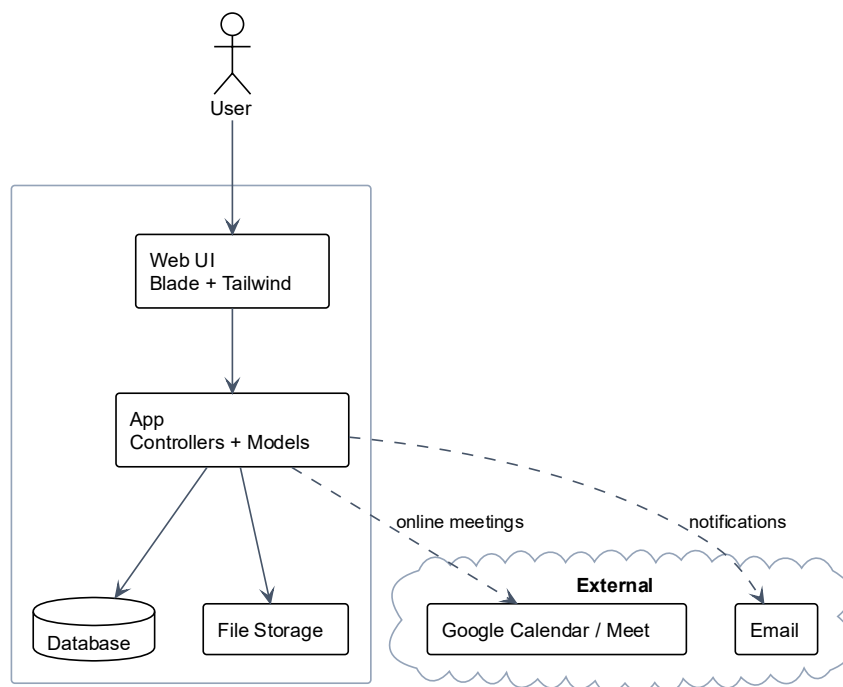
Capability	Student	Supervisor	Admin
Self-register	Yes	Yes	No (promoted only)
Create / edit / submit proposal	Yes	No	No
Review & decide proposal	No	Yes	No
View own / supervised thesis	Own only	Supervised only	All (admin panel)
Submit final thesis	Yes	No	No
Create / edit / delete milestones & tasks	No	Yes (while open)	No
Complete milestone / update own task status	Yes	No	No
Schedule / edit / delete meetings	No	Yes	No
Post public comment	Yes	Yes	No
Post private (supervisor-only) note	No	Yes	No
Upload documents / new versions	Yes	Yes	No
Download documents	Own thesis	Supervised thesis	No admin route
Override thesis status	No	No	Yes
Manage users (role/department/active)	No	No	Yes
Manage departments (CRUD)	No	No	Yes
Connect Google Calendar	No	Yes	No

Functional Requirements

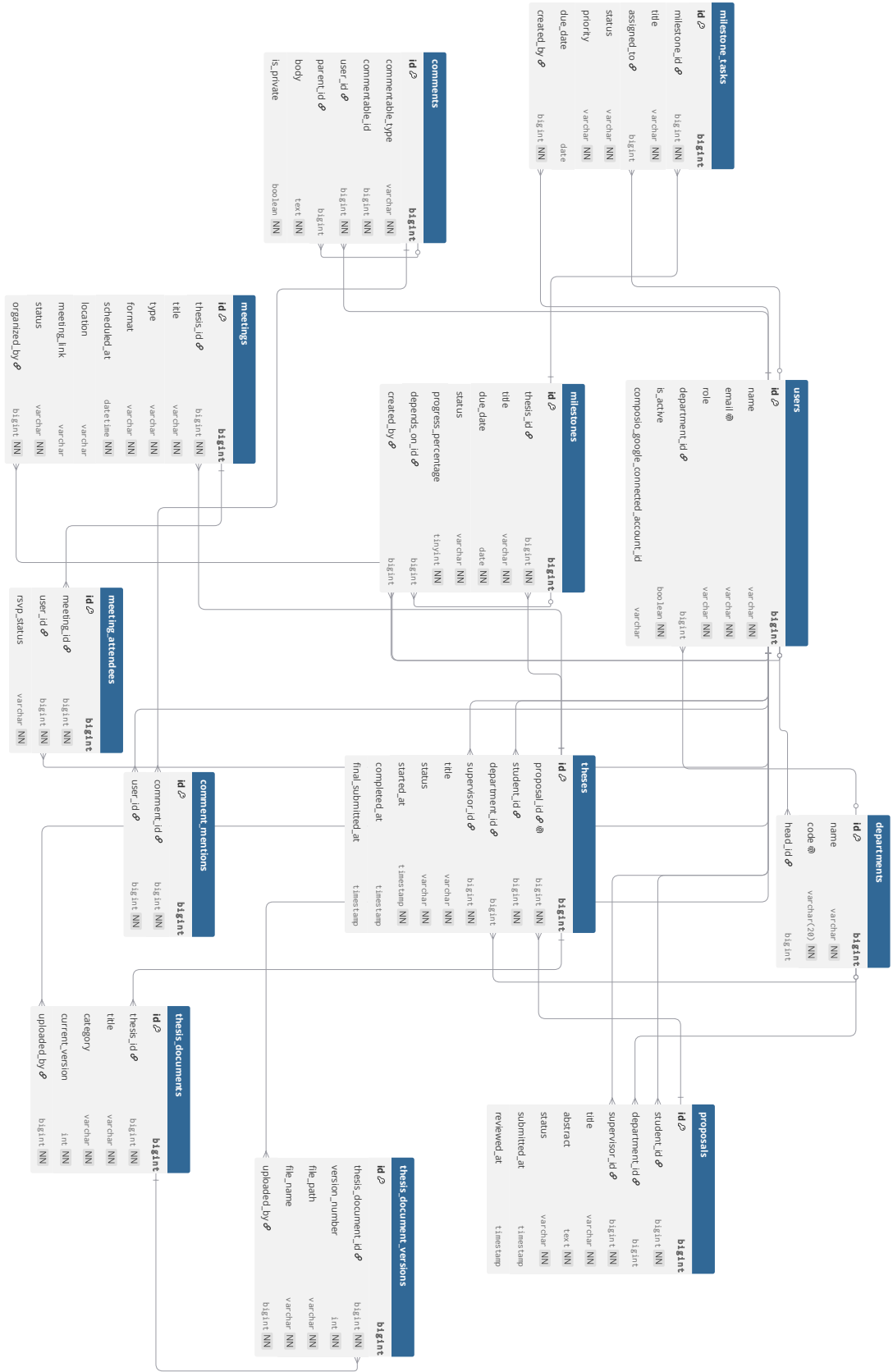
- FR1 - Allow a Student to create, edit (while Draft/Revision-Requested), and submit a research proposal to a chosen active Supervisor.
- FR2 - Allow a Supervisor to review a submitted proposal and record Approve / Reject / Request-Revision, with mandatory notes for the latter two.
- FR3 - On approval, automatically create a Thesis linked 1:1 to the approved proposal, copying student, supervisor, and department.
- FR4 - Allow a Supervisor to define Milestones (due dates, optional dependency) and Tasks under each milestone, assigned to the student.
- FR5 - Automatically recompute a milestone's progress percentage from its tasks, and block completion unless all tasks are done and any dependency is itself complete.
- FR6 - Support thesis document uploads under categories (Proposal, Chapter, Appendix, Final, Other) as immutable, SHA-256-checksummed versions.
- FR7 - Allow a Supervisor to schedule a Meeting and auto-provision a Google Calendar event with a Meet link when Google is connected and the meeting is Online.
- FR8 - Support threaded comments (one level of replies), @mention parsing scoped to the thesis's two participants, and supervisor-only private notes.
- FR9 - Gate final thesis submission on the presence of an uploaded Final-category document.
- FR10 - Provide role-specific dashboards with live counts (pending reviews, active theses, overdue milestones, totals).
- FR11 - Provide an Admin interface for department CRUD and user role/status management, including confirmed promotion to Admin.

System Design

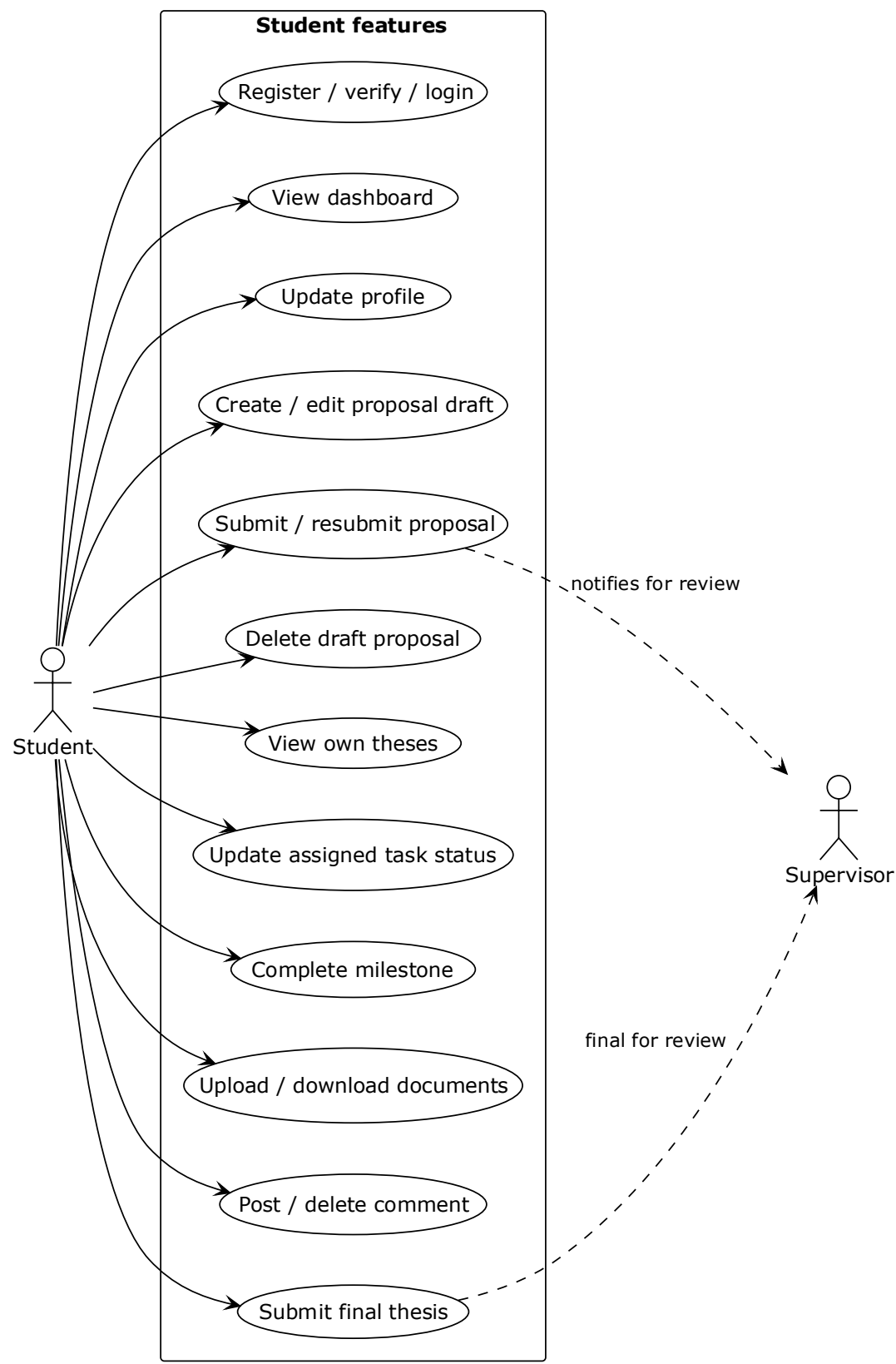
System Architecture



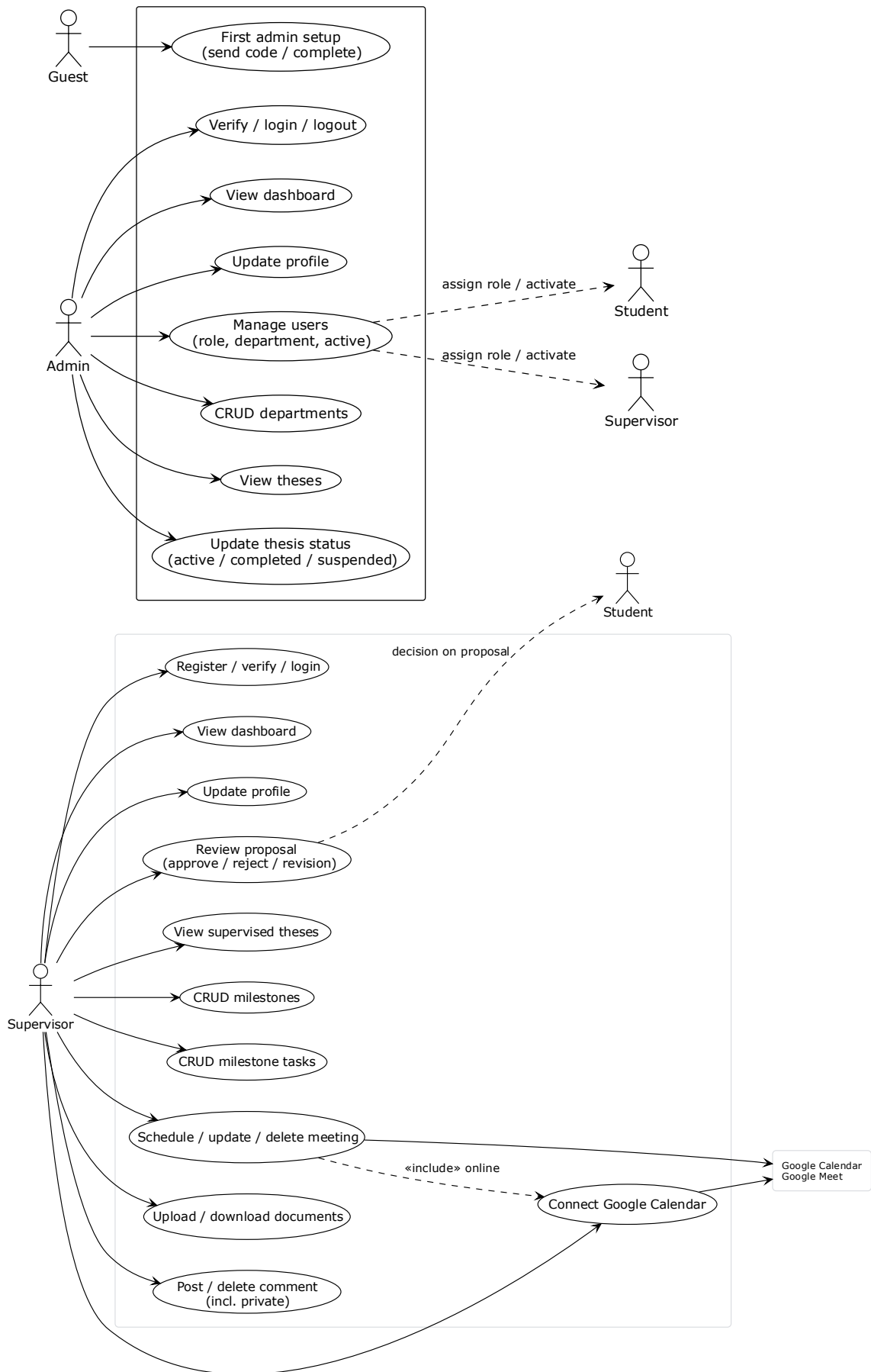
Schema Diagram



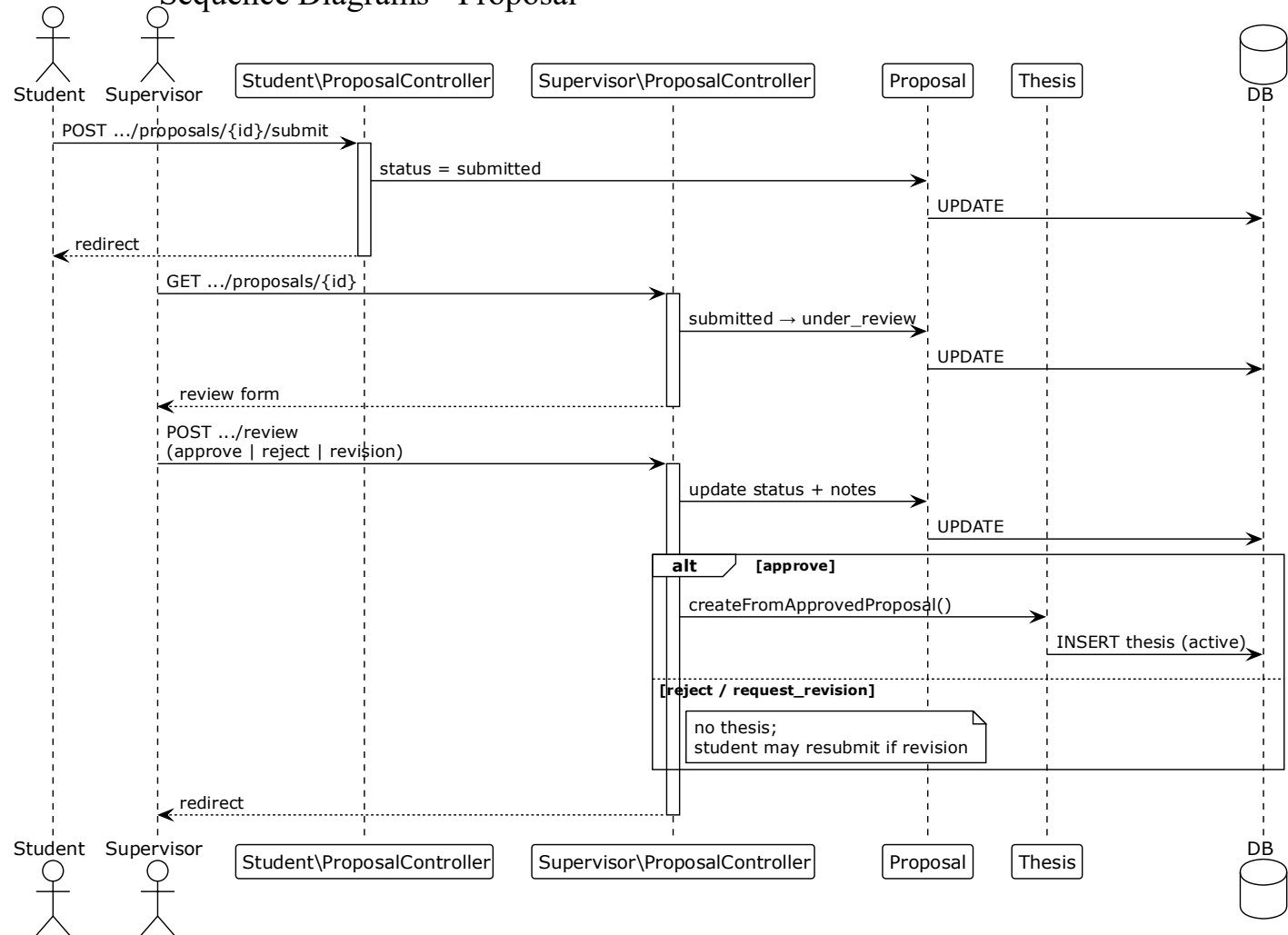
Use Case Diagrams



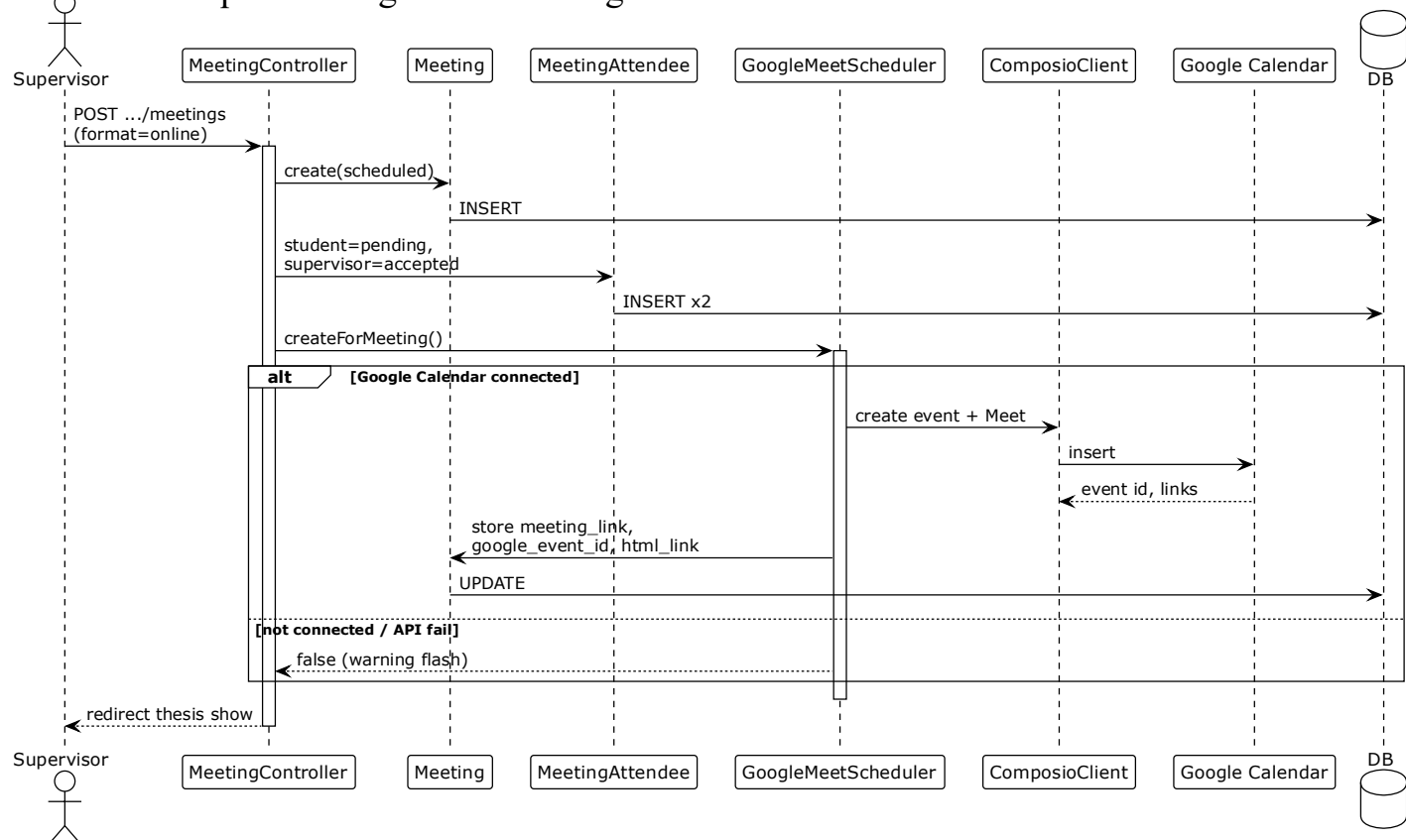
Use Case Diagrams



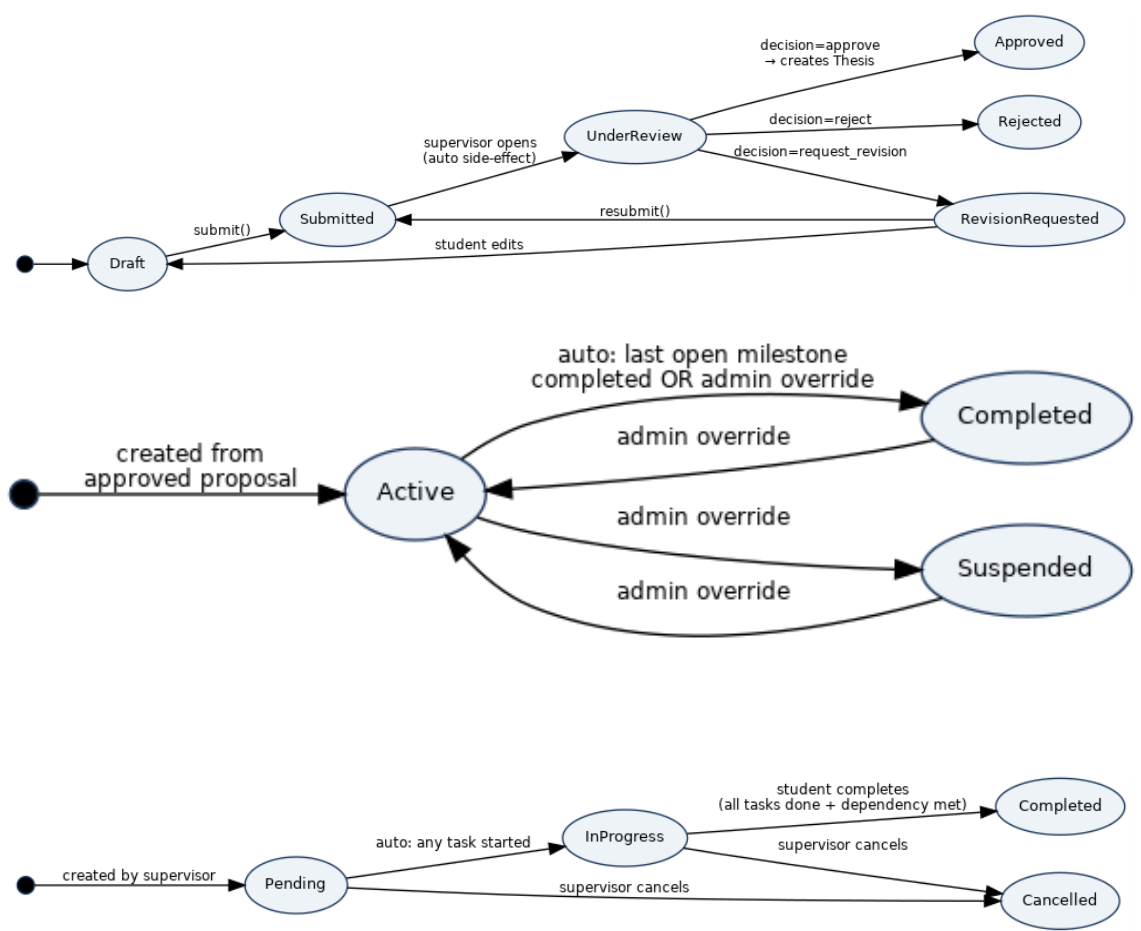
Sequence Diagrams - Proposal



Sequence Diagrams - Meeting



State Diagrams



Tech Stack

Package / Tool	Role
laravel/framework	Core application framework
php	Language runtime
livewire/livewire	wire:navigate SPA-style page transitions only
guzzlehttp/guzzle	HTTP client used by the Composio integration service
vite	Frontend build tool
tailwindcss / @tailwindcss/vite	Utility-first CSS framework
laravel-vite-plugin	Laravel ↔ Vite bridge

Live Application Walkthrough - Admin Profile Setup



Initial administrator setup

Create the first administrator account before anyone else can use the platform.

Authorized setup email

r*****@stud.kuet.ac.bd

Only this address can receive the setup code and become the first admin.

Send setup code

A one-time code will be emailed to the configured address. Codes expire after 60 minutes.

Already have a code? [Complete setup](#)

Setup Code Sent



A setup code has been sent to the configured administrator email.

Complete administrator setup

Enter the setup code from your email and choose administrator credentials.

Administrator email

rahman2207097@stud.kuet.ac.bd

Displayed as r*****@stud.kuet.ac.bd. This cannot be changed during setup.

Setup code

XXXX-XXXX-XXXX-XXXX

Full name

Password

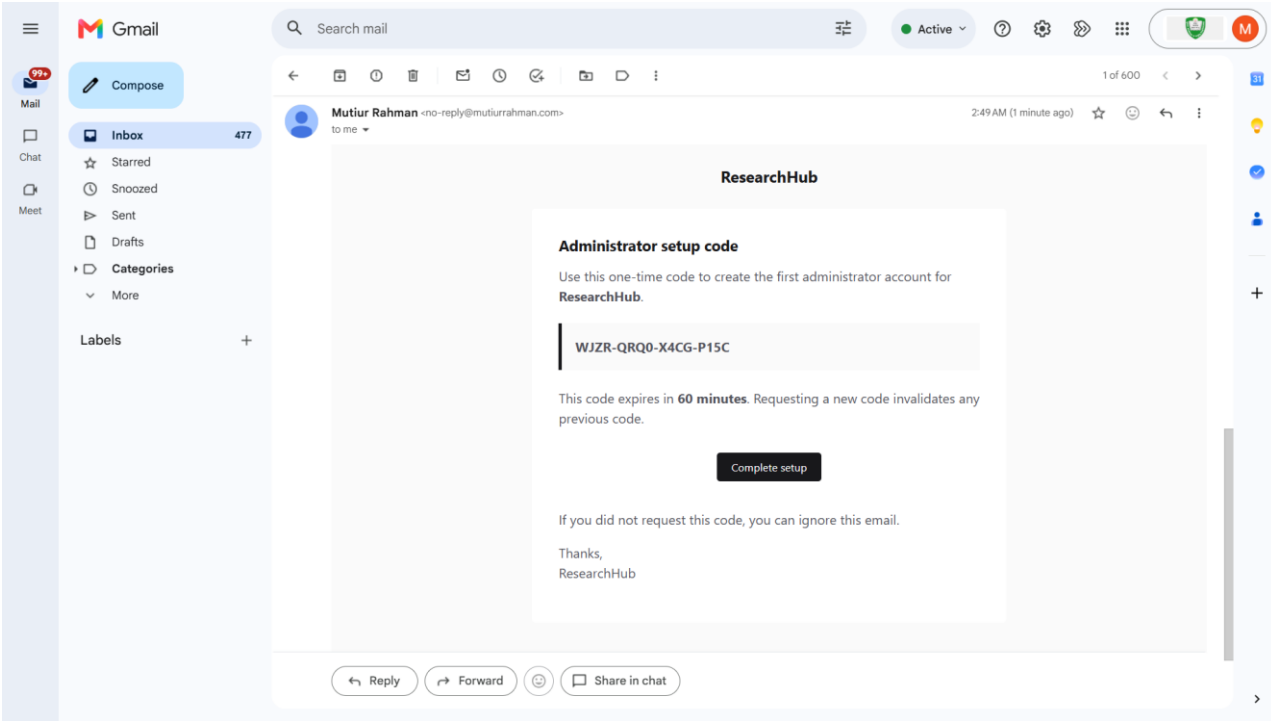
Minimum 8 characters.

Confirm password

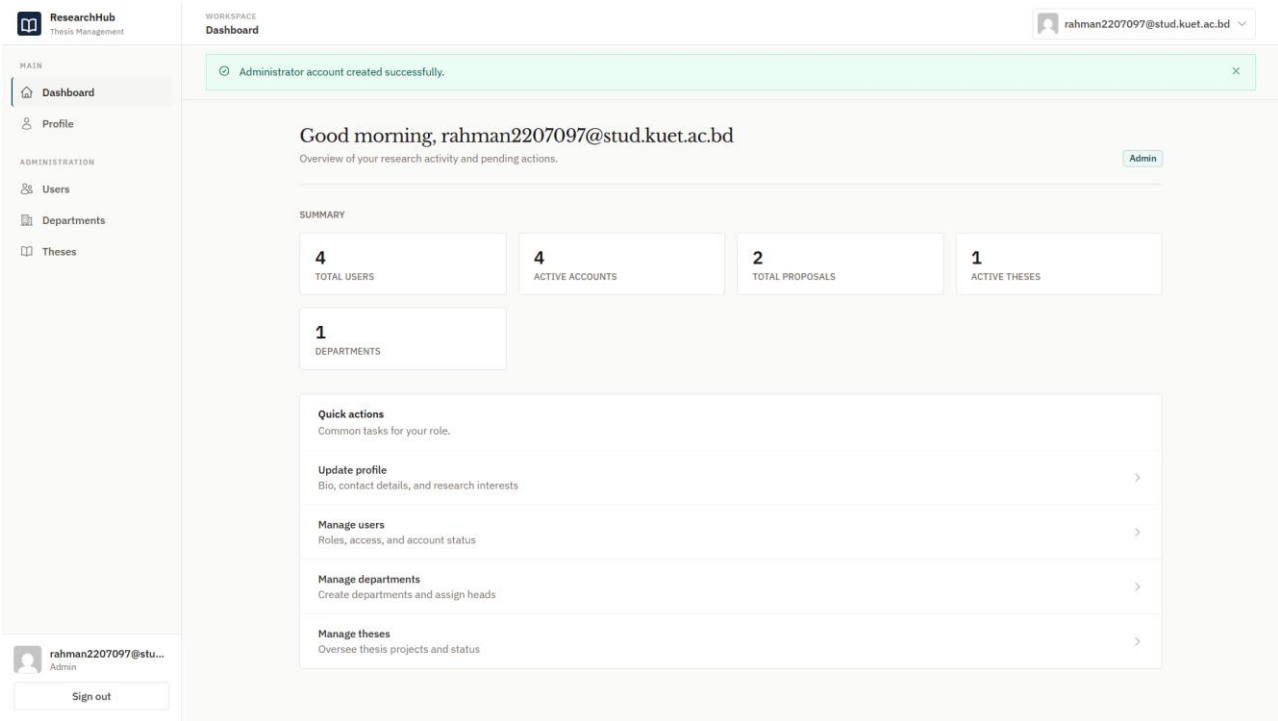
Create administrator

Need a new code? [Request another code](#)

Email Received

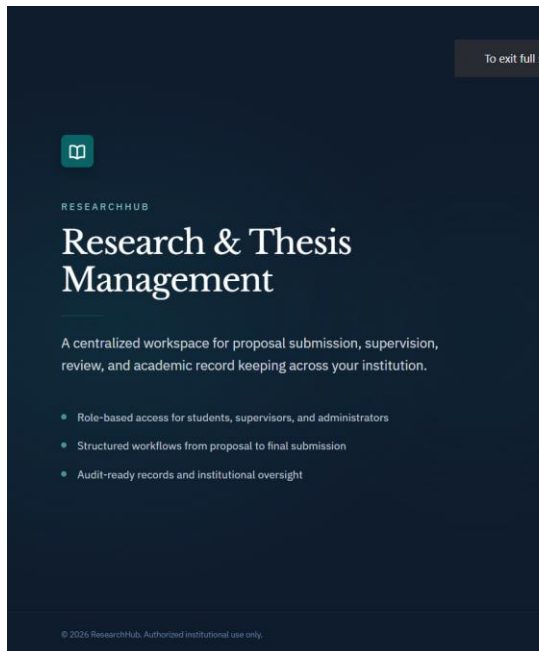


Admin Created



Live Application Walkthrough - Student Profile

Home Page



Get started

Sign in with your institutional account, or create one as a student or supervisor.

[Sign in](#)

[Create account](#)

Students and supervisors can self-register. Administrator accounts are created during platform setup.

Login Page



Sign in

Use your institutional email to access proposals, supervision, and records.

Email address

name@dept.kuet.ac.bd

Password

[Forgot password?](#)

☐ Keep me signed in on this device

[Sign in](#)

No account? [Create account](#)

Student Dashboard

ResearchHub

Thesis Management

WORKSPACE

Dashboard

Mutiur Rahman

MAIN

Dashboard

Profile

Proposals

Theses

Good morning, Mutiur Rahman

Overview of your research activity and pending actions.

Student

SUMMARY

1

MY PROPOSALS

0

PENDING REVIEW

0

MILESTONES DUE

1

ACTIVE THESES

Quick actions

Common tasks for your role.

Update profile

Bio, contact details, and research interests

>

My proposals

Create drafts and submit for supervisor review

>

My theses

Track active research projects

>

Mutiur Rahman

Student

Sign out

Student Profile

ResearchHub

Thesis Management

WORKSPACE

Profile

Mutiur Rahman

MAIN

Dashboard

Profile

Proposals

Theses

Profile

Personal information and research profile.

Photo



Upload photo

JPG, PNG, or WebP. Maximum 2 MB.

Personal information

Full name

Mutiur Rahman

Email address

mutiur5bb@gmail.com

Contact an administrator to change your email.

Role

Student

Department

CSE — Computer Science & Engineering

Contact an administrator to change your department.

Phone number

01711000002

11 digits starting with 01 (e.g. 01712345678).

Mutiur Rahman

Student

Sign out

Proposal Page

ResearchHub

Thesis Management

WORKSPACE

New Proposal

Mutiur Rahman

MAIN

Dashboard

Profile

Proposals

Theses

New proposal

Save a draft first. You can submit it to your supervisor when you are ready.

Proposal details

Describe your research topic and intended approach.

Title

Research title...

Supervisor

Select a supervisor

Abstract

Summarize the research problem, significance, and expected contribution...

Objectives

List the main research objectives...

Methodology

Outline your proposed methods and tools...

Cancel

Save draft

Thesis Page -1

ResearchHub

Thesis Management

WORKSPACE

AI-Powered Thesis Plagiarism Detection for Bengali Text

Mutiur Rahman

MAIN

Dashboard

Profile

Proposals

Theses

Active

AI-Powered Thesis Plagiarism Detection for Bengali Text

Supervisor: Dr. Mutiur Rahman · CSE

Back to List

Approved Proposal

This thesis project was created from your approved proposal.

View Original Proposal—

Abstract

Existing plagiarism tools are optimized for English and often miss paraphrased Bengali academic writing. This research proposes a semantic plagiarism detection pipeline that combines multilingual transformer embeddings with institutional thesis repositories. The goal is to support supervisors with ranked similarity reports that highlight suspected reused passages across Bengali and mixed-language submissions.

Timeline

Milestone schedule from project start to final due date.

May 21, 2026

Literature review co...
Jun 19

Model training and e...
Aug 2

Final thesis draft
Aug 19

Aug 26, 2026

Milestones

Track progress on supervisor-defined deliverables and tasks.

Literature review complete

Completed

Due Jun 19, 2026 100% - 2 tasks

Final Submission

Submit your completed thesis for supervisor review.

Upload a document with category **Final thesis**, then submit it here.

Submit Final Thesis

Timeline

STARTED
May 21, 2026 3:08 PM

Thesis Page -2

ResearchHub

Thesis Management

WORKSPACE

AI-Powered Thesis Plagiarism Detection for Bengali Text

Mutiur Rahman

MAIN

Dashboard

Profile

Proposals

Theses

Mutiur Rahman
Student

Sign out

Milestones

Track progress on supervisor-defined deliverables and tasks.

Literature review complete

Completed

Due Jun 19, 2026 · 100% · 2 tasks

Model training and evaluation

Completed

Due Aug 2, 2026 · 100% · 2 tasks

Final thesis draft

Pending

Due Aug 19, 2026 · 0% · 1 task

Documents

Upload thesis files with automatic version tracking.

Upload Document

Last Chapter

Chapter

v1.0

Download Latest

New Version

Latest: Git_GriHub_Lab_Report.pdf · 533.9 KB · Jul 20, 2026

Chapter 1 - Introduction

Chapter

v1.0

Download Latest

New Version

Initial chapter draft covering motivation, problem statement, and research questions.
Latest: chapter-1-introduction.pdf · 694 B · Jul 19, 2026

Final Thesis Draft

Final thesis

v2.0

Download Latest

New Version

Working final draft for supervisor review before formal submission.
Latest: chapter-1-introduction.pdf · 694 B · Jul 19, 2026

Version History (2)

Meetings

Schedule supervision sessions and record agendas and minutes.

No meetings scheduled yet.

Discussion

Thesis Page -3

ResearchHub

Thesis Management

WORKSPACE

AI-Powered Thesis Plagiarism Detection for Bengali Text

Mutiur Rahman

MAIN

Dashboard

Profile

Proposals

Theses

Mutiur Rahman
Student

Sign out

version history (2)

v2.0 · chapter-1-introduction.pdf

694 B · Dr. Mutiur Rahman · Jul 19, 2026 11:17 PM

Download

v1.0 · final-thesis-draft.pdf

670 B · Mutiur Rahman · Jul 19, 2026 3:08 PM

First complete draft

Download

Meetings

Schedule supervision sessions and record agendas and minutes.

No meetings scheduled yet.

Discussion

Project thread for updates and feedback.

New comment

Type: @ or pick someone below.

Mention

@ Dr. Mutiur Rahman

Share an update or ask a question.

Post Comment

Dr. Mutiur Rahman

2 hours ago

Mentioned: Mutiur Rahman

@Mutiur Rahman

Reply

Dr. Mutiur Rahman

11 hours ago

Good progress on the literature review. Please add more Bengali-language baseline comparisons in the next draft.

Reply

Mutiur Rahman

11 hours ago

Noted. I will include BanglaBERT and a lexical TF-IDF baseline in the evaluation table.

Delete

Live Application Walkthrough - Supervisor Profile

Dashboard

The screenshot shows the ResearchHub Supervisor Dashboard for Dr. Mutiur Rahman. The interface includes a sidebar with navigation links: Dashboard, Profile, Reviews, and Theses. The main content area displays a greeting, a summary of key metrics, and a list of quick actions.

ResearchHub
Thesis Management

WORKSPACE
Dashboard

Dr. Mutiur Rahman

Good morning, Dr. Mutiur

Overview of your research activity and pending actions.

SUMMARY

Count	Category
2	PENDING REVIEWS
4	SUPERVISED PROPOSALS
0	OVERDUE MILESTONES
1	ACTIVE PROJECTS

Quick actions
Common tasks for your role.

- Update profile**
Bio, contact details, and research interests
- Review proposals**
Approve, reject, or request revisions
- Supervised theses**
View active student research projects

Dr. Mutiur Rahman
Supervisor

Sign out

Composio Connection for Meeting

The screenshot shows a Composio connection overlay for Google Calendar. The overlay includes the Composio logo, a date indicator (31), and a message stating "Composio wants to connect to your Google Calendar". It also indicates that the user is being taken to Google Calendar and should authenticate to continue. A "Connecting..." button is visible at the bottom.

Composio wants to connect to your Google Calendar

Taking you to Google Calendar
Please authenticate to continue

Connecting...

Secured by Composio

Proposal Review Page

ResearchHub

Thesis Management

MAIN

Dashboard

Profile

Reviews

Theses

Dr. Mutiur Rahman

Supervisor

Sign out

WORKSPACE

Proposal Reviews

Dr. Mutiur Rahman

Proposal reviews

Review proposals submitted by your students.

Status

Pending review

Apply

TITLE	STUDENT	STATUS	SUBMITTED	ACTIONS
Suuuuuuuuuu	Arif Mahmud helloworldkuet@gmail.com	Under review	2 hours ago	Review
Blockchain-Based Academic Credential Verification korbona	Arif Mahmud helloworldkuet@gmail.com	Under review	2 hours ago	Review

Proposal Response Page

ResearchHub

Thesis Management

MAIN

Dashboard

Profile

Reviews

Theses

Dr. Mutiur Rahman

Supervisor

Sign out

WORKSPACE

Review Proposal

Dr. Mutiur Rahman

Under review

Back to list

Blockchain-Based Academic Credential Verification korbona

Student: Arif Mahmud (helloworldkuet@gmail.com)

Abstract

Academic institutions still verify certificates through slow, paper-based channels that are easy to forge. This proposal introduces a decentralized verification workflow that stores credential hashes on a permissioned blockchain and exposes a public verification portal for employers and universities. The system aims to reduce verification time while preserving institutional control over credential issuance.

Objectives

1. Design a credential schema suitable for undergraduate and postgraduate certificates.

2. Implement smart-contract based issuance and revocation on an Ethereum testnet.

3. Build a Laravel verification API and portal for third-party checks.

4. Evaluate security, latency, and operational cost against current manual verification.

Methodology

1. Requirements analysis with the exam controller office.

2. Smart-contract development and unit testing on a public testnet.

3. Backend integration using Laravel for issuer workflows and audit logs.

4. Controlled pilot with synthetic credentials and security review.

Decision

Approve, reject, or request revisions.

Decision

Select decision

Select decision

Approve

Request revision

Reject

Submit decision

Timeline

SUBMITTED

Jul 19, 2026 11:52 PM

Thesis Supervise Page -1

ResearchHub

Thesis Management

MAIN

Dashboard

Profile

Reviews

Theses

Dr. Mutiur Rahman

Supervisor

Sign out

WORKSPACE

AI-Powered Thesis Plagiarism Detection for Bengali Text

Dr. Mutiur Rahman

Active

Back to List

Source Proposal

This thesis was created when you approved the student's proposal.

View Original Proposal--

Timeline

STARTED

May 21, 2026 3:08 PM

Abstract

Existing plagiarism tools are optimized for English and often miss paraphrased Bengali academic writing. This research proposes a semantic plagiarism detection pipeline that combines multilingual transformer embeddings with institutional thesis repositories. The goal is to support supervisors with ranked similarity reports that highlight suspected reused passages across Bengali and mixed-language submissions.

Timeline

Milestone schedule from project start to final due date.

May 21, 2026

Jun 19

Aug 2

Aug 19

Aug 26, 2026

Milestones

Define deliverables, assign tasks, and set dependencies.

Literature review complete

Completed

Due Jun 19, 2026 100% - 2 tasks

Thesis Supervise Page -2

ResearchHub

Thesis Management

MAIN

Dashboard

Profile

Reviews

Theses

Dr. Mutiur Rahman

Supervisor

Sign out

WORKSPACE

AI-Powered Thesis Plagiarism Detection for Bengali Text

Dr. Mutiur Rahman

Literature review complete

Completed

Due Jun 19, 2026 100% - 2 tasks

Survey plagiarism detection systems, Bengali NLP resources, and evaluation metrics.

Completed Jun 28, 2026

Progress 100%

Task

Priority

Status

Actions

Collect related papers

Gather at least 25 peer-reviewed papers on semantic plagiarism and Bengali NLP.

Due Jun 7, 2026

High

Completed

Edit

Delete

Write literature summary

Summarize gaps in English-centric tools for Bengali academic text.

Due Jun 19, 2026

Medium

Completed

Edit

Delete

Add Task

Model training and evaluation

Completed

Due Aug 2, 2026 100% - 2 tasks

Train embedding model, run benchmarks, and document results.

Depends on: Literature review complete Completed Jul 20, 2026

Progress 100%

Task

Priority

Status

Actions

Prepare training dataset

Clean, tokenize, and split the Bengali thesis corpus.

Due Jul 12, 2026

High

Completed

Edit

Delete

Run baseline experiments

Compare TF-IDF, multilingual MiniLM, and fine-tuned embeddings.

Due Jul 26, 2026

Urgent

Completed

Edit

Delete

Add Task

Task title

e.g. Draft introduction section.

Thesis Supervise Page -3

ResearchHub

Thesis Management

MAIN

Dashboard

Profile

Reviews

Theses

Dr. Mutiur Rahman

Supervisor

Sign out

WORKSPACE

AI-Powered Thesis Plagiarism Detection for Bengali Text

Dr. Mutiur Rahman

Add Task

Task title

e.g. Draft introduction section.

Priority

Low

Due date (optional)

mm/dd/yyyy

Description (optional)

Add Task

Final thesis draft

Pending

Due Aug 19, 2026 · 0% · 1 task

Add Milestone

Documents

Upload Document

Upload thesis files with automatic version tracking.

Last Chapter

Chapter

v1.0

Download Latest

New Version

Latest: Git_Github_Lab_Report.pdf · 533.9 KB · Jul 20, 2026

Chapter 1 - Introduction

Chapter

v1.0

Download Latest

New Version

Initial chapter draft covering motivation, problem statement, and research questions.
Latest: chapter-1-introduction.pdf · 694 B · Jul 19, 2026

Final Thesis Draft

Final thesis

v2.0

Download Latest

New Version

Working final draft for supervisor review before formal submission.
Latest: chapter-1-introduction.pdf · 694 B · Jul 19, 2026

Version History (2)

Meetings

Thesis Supervise Page -4

ResearchHub

Thesis Management

MAIN

Dashboard

Profile

Reviews

Theses

Dr. Mutiur Rahman

Supervisor

Sign out

WORKSPACE

AI-Powered Thesis Plagiarism Detection for Bengali Text

Dr. Mutiur Rahman

Version History (2)

Meetings

Schedule supervision sessions and record agendas and minutes.

Schedule Meeting

No meetings scheduled yet.

Discussion

Project thread for updates and feedback.

New comment

Type @ or pick someone below.

Mention @ Mutiur Rahman

@

Mutiur Rahman

mutiur5bb@gmail.com

Dr. Mutiur Rahman

2 hours ago

Mentioned: Mutiur Rahman

@Mutiur Rahman

Reply

Delete

Dr. Mutiur Rahman

11 hours ago

Good progress on the literature review. Please add more Bengali-language baseline comparisons in the next draft.

Reply

Delete

Mutiur Rahman

11 hours ago

Noted. I will include BanglaBERT and a lexical TF-IDF baseline in the evaluation table.

Delete

Dr. Mutiur Rahman

11 hours ago

Private

Private note: push the student to document dataset licensing constraints before the final chapter.

Reply

Delete

Meeting Manage - 1

ResearchHub

Thesis Management

MAIN

Dashboard

Profile

Reviews

Theses

Dr. Mutiur Rahman

Supervisor

Sign out

WORKSPACE

AI-Powered Thesis Plagiarism Detection for Bengali Text

Dr. Mutiur Rahman

Final Thesis Draft

Final thesis

v2.0

Download Latest

New Version

Working final draft for supervisor review before formal submission.
Latest: chapter-1-introduction.pdf - 694 B - Jul 19, 2026

> Version History (2)

Meetings

Schedule supervision sessions and record agendas and minutes.

Schedule Meeting

Connect Google Calendar in your profile to auto-create Meet links for online meetings.

Title

e.g. Weekly supervision check-in.

Type

Supervision

Format

Online

Video call via Google Meet

In person

Requires a physical location

Date & time (Asia/Dhaka)

07/20/2026 04:30 PM

Duration (minutes)

60

Agenda (optional)

Topics to cover in this meeting.

Notes (optional)

Additional context.

ScheduleCancel

No meetings scheduled yet.

Discussion

Project thread for updates and feedback.

Meeting Manage - 2

ResearchHub

Thesis Management

MAIN

Dashboard

Profile

Reviews

Theses

Dr. Mutiur Rahman

Supervisor

Sign out

WORKSPACE

AI-Powered Thesis Plagiarism Detection for Bengali Text

Dr. Mutiur Rahman

Upload Document

Upload thesis files with automatic version tracking.

Last Chapter

Chapter

v1.0

Download Latest

New Version

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Chapter 1 - Introduction

Chapter

v1.0

Download Latest

New Version

Initial chapter draft covering motivation, problem statement, and research questions.
Latest: chapter-1-introduction.pdf - 694 B - Jul 19, 2026

Final Thesis Draft

Final thesis

v2.0

Download Latest

New Version

Working final draft for supervisor review before formal submission.
Latest: chapter-1-introduction.pdf - 694 B - Jul 19, 2026

> Version History (2)

Meetings

Schedule supervision sessions and record agendas and minutes.

Schedule Meeting

Weekly Supervision

Supervision

Online

Jul 20, 2026 4:30 PM - 60 min

EditDelete

Mutiur Rahman

Dr. Mutiur Rahman

Discussion

Project thread for updates and feedback.

New comment

Type: @ or pick someone below.

Mention

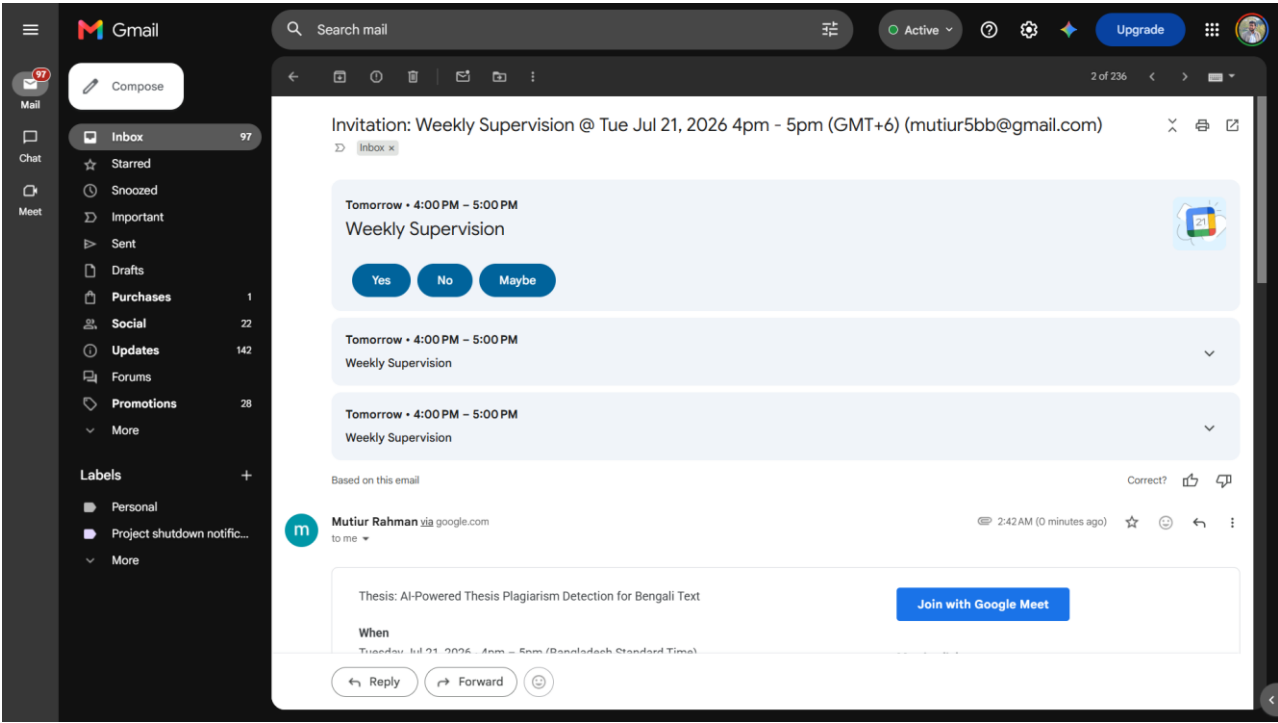
@ Mutiur Rahman

Share an update or ask a question.

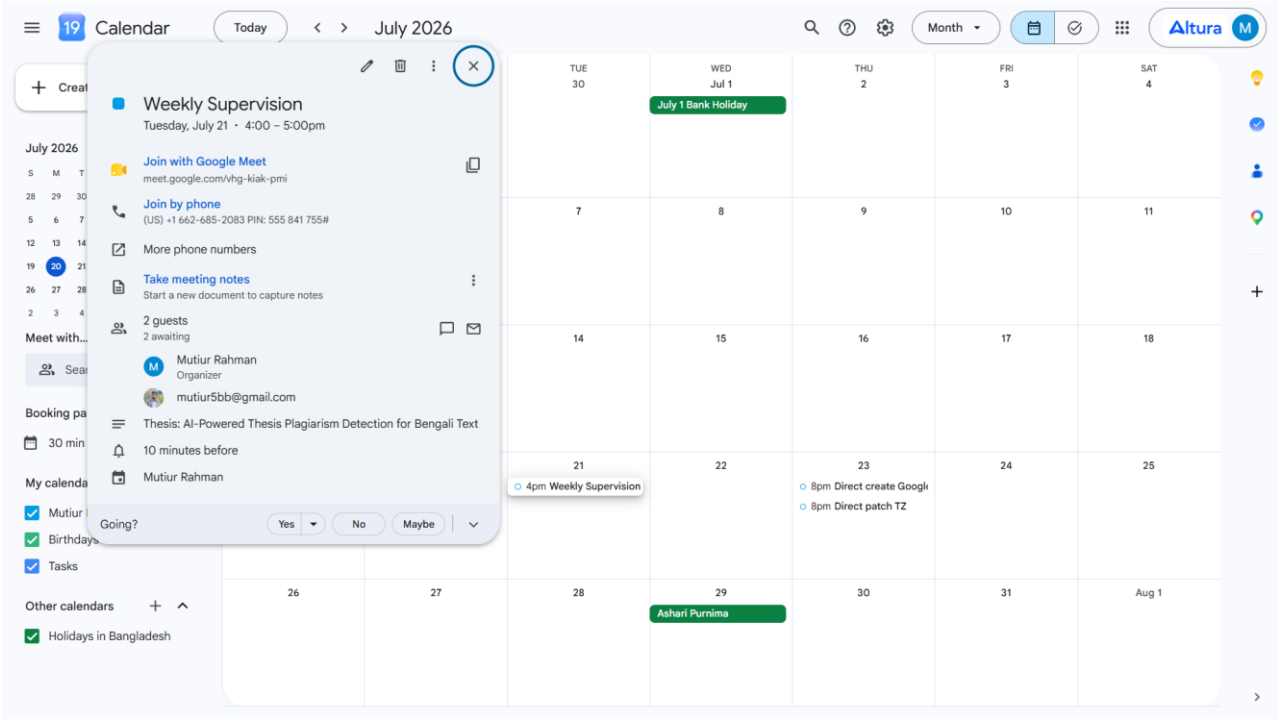
☐ Private note (visible to supervisors only)

Post Comment

Meeting Manage - 3



Meeting Manage - 4



Meeting Manage - 5

